Chapter 1

Mechanics

1.1 Point-kinetics in a fixed coordinate system

1.1.1 Definitions

The position $\vec{r}$, the velocity $\vec{v}$ and the acceleration $\vec{a}$ are defined by: $\vec{r} = (x, y, z)$, $\vec{v} = (\dot{x}, \dot{y}, \dot{z})$, $\vec{a} = (\ddot{x}, \ddot{y}, \ddot{z})$. The following holds:

$$s(t) = s_0 + \int |\vec{v}(t)| dt \ ; \quad \vec{r}(t) = \vec{r}_0 + \int \vec{v}(t) dt \ ; \quad \vec{v}(t) = \vec{v}_0 + \int \vec{a}(t) dt$$

When the acceleration is constant this gives: $v(t) = v_0 + at$ and $s(t) = s_0 + v_0 t + \frac{1}{2} a t^2$.
For the unit vectors in a direction $\parallel$ to the orbit $\vec{e}_\mathrm{t}$ and parallel to it $\vec{e}_\mathrm{n}$ holds:

$$\vec{e}_\mathrm{t} = \frac{\vec{v}}{|\vec{v}|} = \frac{d\vec{r}}{ds} \quad \dot{\vec{e}}_\mathrm{t} = \frac{v}{\rho} \vec{e}_\mathrm{n} \ ; \quad \vec{e}_\mathrm{n} = \frac{\dot{\vec{e}}_\mathrm{t}}{|\dot{\vec{e}}_\mathrm{t}|}$$

For the *curvature* k and the *radius of curvature* ρ holds:

$$\vec{k} = \frac{d\vec{e}_\mathrm{t}}{ds} = \frac{d^2\vec{r}}{ds^2} = \left| \frac{d\varphi}{ds} \right| \ ; \quad \rho = \frac{1}{|k|}$$

1.1.2 Polar coordinates

Polar coordinates are defined by: $x = r\cos(\theta)$, $y = r\sin(\theta)$. So, for the unit coordinate vectors holds: $\vec{e}_r = \theta \vec{e}_\theta$, $\vec{e}_\theta = -\theta \vec{e}_r$.

The velocity and the acceleration are derived from: $\vec{r} = r\vec{e}_r$, $\vec{v} = \dot{r}\vec{e}_r + r\dot{\theta}\vec{e}_\theta$, $\vec{a} = (\ddot{r} - r\dot{\theta}^2)\vec{e}_r + (2\dot{r}\dot{\theta} + r\ddot{\theta})\vec{e}_\theta$.

1.2 Relative motion

For the motion of a point D w.r.t. a point Q holds: $\vec{r}_\mathrm{D} = \vec{r}_\mathrm{Q} + \frac{\vec{\omega} \times \vec{v}_\mathrm{Q}}{\omega^2}$ with $\vec{\mathrm{QD}} = \vec{r}_\mathrm{D} - \vec{r}_\mathrm{Q}$ and $\omega = \dot{\theta}$.

Further holds: $\alpha = \dot{\theta}$. $'$ means that the quantity is defined in a moving system of coordinates. In a moving system holds:
$\vec{v} = \vec{v}_\mathrm{Q} + \vec{v}\,' + \vec{\omega} \times \vec{r}\,'$ and $\vec{a} = \vec{a}_\mathrm{Q} + \vec{a}\,' + \vec{\alpha} \times \vec{r}\,' + 2\vec{\omega} \times \vec{v} - \vec{\omega} \times (\vec{\omega} \times \vec{r}\,')$
with $|\vec{\omega} \times (\vec{\omega} \times \vec{r}\,')| = \omega^2 \vec{r}_n\,'$

1.3 Point-dynamics in a fixed coordinate system

1.3.1 Force, (angular)momentum and energy

Newton's 2nd law connects the force on an object and the resulting acceleration of the object:

$$\vec{F}(\vec{r}, \vec{v}, t) = m\vec{a} = \frac{d\vec{p}}{dt} \quad \text{, where the } \textit{momentum} \text{ is given by: } \ \vec{p} = m\vec{v}$$

Newton's 3rd law is: $\vec{F}_\mathrm{action} = -\vec{F}_\mathrm{reaction}$.

For the power P holds: $P = \dot{W} = \vec{F} \cdot \vec{v}$. For the total energy W, the kinetic energy T and the potential energy U holds: $W = T + U$; $T = -U$ with $T = \frac{1}{2}mv^2$.

The *kick* $\vec{S}$ is given by: $\vec{S} = \Delta\vec{p} = \int \vec{F} dt$

The work A, delivered by a force, is $A = \int_1^2 \vec{F} \cdot d\vec{s} = \int_1^2 F \cos(\alpha) ds$

The torque $\vec{\tau}$ is related to the angular momentum $\vec{L}$: $\vec{\tau} = \dot{\vec{L}} = \vec{r} \times \vec{F}$; and $\vec{L} = \vec{r} \times \vec{p} = m\vec{v} \times \vec{r}$, $|\vec{L}| = mr^2\omega$. The following holds:

$$\tau = -\frac{\partial U}{\partial \theta}$$

So, the conditions for a mechanical equilibrium are: $\sum \vec{F}_i = 0$ and $\sum \vec{\tau}_i = 0$.

The *force of friction* is usually proportional with the force perpendicular to the surface, except when the motion starts, when a threshold has to be overcome: $F_{\text{fric}} = f \cdot F_{\text{norm}} \cdot \vec{e}_t$.

1.3.2 Conservative force fields

A conservative force can be written as the gradient of a potential: $\vec{F}_{\text{cons}} = -\vec{\nabla} U$. From this follows that $\text{rot} F = 0$. For such a force field also holds:

$$\oint \vec{F} \cdot d\vec{s} = 0 \;\Rightarrow\; U = U_0 - \int_{r_0}^{r_1} \vec{F} \cdot d\vec{s}$$

So the work delivered by a conservative force field depends not on the followed trajectory but only on the starting and ending points of the motion.

1.3.3 Gravitation

The Newtonian law of gravitation is (in GRT one also uses κ instead of G):

$$\vec{F}_{\text{g}} = -G\frac{m_1 m_2}{r^2}\vec{e}_r$$

The gravitationpotential is then given by $V = -Gm/r$. From Gauss law then follows: $\nabla^2 V = 4\pi G\varrho$.

1.3.4 Orbital equations

From the equations of Lagrange for ϕ, conservation of angular momentum can be derived:

$$\frac{\partial \mathcal{L}}{\partial \phi} = \frac{\partial V}{\partial \phi} = 0 \Rightarrow \frac{d}{dt}(mr^2\phi) = 0 \Rightarrow L_z = mr^2\phi = \text{constant}$$

For the radius as a function of time can be found that:

$$\left(\frac{dr}{dt}\right)^2 = \frac{2(W-V)}{m} - \frac{L^2}{m^2r^2}$$

The angular equation is then:

$$\phi - \phi_0 = \int_0^r \left[\frac{mr^2}{L}\sqrt{\frac{2(W-V)}{m} - \frac{L^2}{m^2r^2}}\right]^{-1} dr \overset{r^{-2}\text{field}}{=} \arccos\left(1 + \frac{\frac{1}{r} - \frac{1}{r_0}}{\frac{1}{r_0} + km/L_z^2}\right)$$

if $F = F(r)$: $L =$ constant, if F is conservative: $W =$ constant, if $\vec{F} \perp \vec{v}$ then $\Delta T = 0$ and $U = 0$.

Kepler's equations

In a force field $F = kr^{-2}$, the orbits are conic sections (Kepler's 1st law). The equation of the orbit is:

$$r(\theta) = \frac{\ell}{1 + \varepsilon\cos(\theta - \theta_0)} \ , \quad \text{or:} \ \ x^2 + y^2 = (\ell - \varepsilon x)^2$$

with

$$\ell = \frac{L^2}{G\mu^2 M_{\text{tot}}} \ ; \quad \varepsilon^2 = 1 + \frac{2WL^2}{G^2\mu^3 M_{\text{tot}}^2} = 1 - \frac{\ell}{a} \ ; \quad a = \frac{\ell}{1-\varepsilon^2} = \frac{k}{2W}$$

a is half the length of the long axis of the elliptical orbit in case the orbit is closed. Half the length of the short axis is $b = \sqrt{a\ell}$. ε is the *excentricity* of the orbit. Orbits with an equal a are equally shaped. Now, 5 kinds of orbits are possible:

1. $k < 0$ and $\varepsilon = 0$: a circle.
2. $k < 0$ and $0 < \varepsilon < 1$: an ellipse.
3. $k < 0$ and $\varepsilon = 1$: a parabole.
4. $k < 0$ and $\varepsilon > 1$: a hyperbole, curved towards the center of force.
5. $k > 0$ and $\varepsilon > 1$: a hyperbole, curved away of the center of force.

Other combinations are not possible: the total energy in a repulsive force field is always positive so $\varepsilon > 1$.

If the surface between the orbit walked thru between t_1 and t_2 and the focus C around which the planet moves is $A(t_1, t_2)$, Kepler's 2nd law is

$$A(t_1, t_2) = \frac{L_{\text{C}}}{2m}(t_2 - t_1)$$

Kepler's 3rd law is, with T the period and M_{tot} the total mass of the system:

$$\frac{T^2}{a^3} = \frac{4\pi^2}{GM_{\text{tot}}}$$

1.3.5 The virial theorem

The virial theorem for one particle is:

$$\langle m\vec{v}\cdot\vec{r}\rangle = 0 \Rightarrow \langle T\rangle = -\tfrac{1}{2}\left\langle \vec{F}\cdot\vec{r}\right\rangle = \tfrac{1}{2}\left\langle r\frac{dU}{dr}\right\rangle = \tfrac{1}{2}n\langle U\rangle \ \text{ if } U = -\frac{k}{r^n}$$

The virial theorem for a collection of particles is:

$$\langle T\rangle = -\tfrac{1}{2}\left\langle \sum_{\text{particles}} \vec{F}_i\cdot\vec{r}_i + \sum_{\text{pairs}} \vec{F}_{ij}\cdot\vec{r}_{ij}\right\rangle$$

These propositions can also be written as: $2E_{\text{kin}} + E_{\text{pot}} = 0$.

1.4 Point dynamics in a moving coordinate system

1.4.1 Apparent forces

The total force in a moving coordinate system can be found by subtracting the apparent forces from the forces working in the reference frame: $F' = F - F_{\text{app}}$. The different apparent forces are given by:

1. Transformation of the origin: $F_{\text{or}} = -m\vec{a}_o$